

Career Objective

Electronics and Communication Engineering graduate with hands-on skills in **Python, SQL, PowerBI, Data Visualization, ML & DL, LLMs, RAG, and Fine-Tuning**. Strong foundation in data analysis and Generative AI, with an interest in applying analytical and AI techniques to develop **data-driven, intelligent solutions** and contribute effectively to technology-focused roles.

Educational credentials

B.E Electronics and Communication Engineering | CGPA : 7.93

2022 - 26

RVS College of Engineering, Dindigul

Internships

Popular Systems - Embedded System & IoT Intern

Summer Internship - 1 month (Aug 2025)

- Implemented and **tested microcontroller- and sensor-based solutions**, gaining practical experience in hardware interfacing, real-time systems, and IoT development.

CodeAlpha - Full stack Development Intern

Virtual Internship - 1 month (Jan 2026)

- Built application components across frontend, backend, and database layers, applying full-stack development concepts to develop and integrate functional web features
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Skills

- Programming** : Python, HTML, CSS, JavaScript
 - AI/ML**: Machine Learning, Deep Learning, NLP
 - Generative AI**: LLMs, RAG, Prompt Engineering, Fine-Tuning
 - Query Language** : SQL, Power Query, DAX Query
 - Data Analysis** : Data Visualization & Exploratory Data Analysis
 - Core Skills** : Embedded C, Sensor Interfacing, ADAS Designing
 - Tools & Software** : MySQL, Power BI, GitHub
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Projects

1. **Intelligent Medical Knowledge Assistant | Generative AI & RAG**

Aug 2026

- Developed a RAG-based medical assistant using **LangChain, and Merck Manuals** to provide evidence-based responses to clinical queries, Processed 4,000+ pages of medical knowledge.
- Evaluated Prompt Engineering vs. RAG, achieving approximately **4.5/5 groundedness & relevance** with the RAG approach.

2. **Food & Health Nutrition Analytics | SQL & Power BI**

Aug 2026

 - Analyzed and cleaned a nutrition-focused dataset using MySQL, applying filtering and aggregation techniques to identify key **food, health, and nutritional insights**.
 - Transformed SQL-based analysis into an interactive Power BI dashboard with KPIs and visualizations to highlight **nutritional trends and patterns**.
3. **Social Media Analytics & Insights | Python**

Jun 2026

 - Parses post metadata (name, position, state) directly from labels, Maps each state to its region.
 - Runs sentiment analysis with **NLTK's VADER**, classifying posts as **positive/negative/neutral**
 - Computes sentiment quantiles by group, most-common hashtags, and per-hashtag sentiment scores
4. **Intelligent Driver Monitoring & Emergency Control System**

May 2026

 - Developed a real-time driver drowsiness detection system using **Python, OpenCV, Dlib, and image processing**, analyzing Eye Aspect Ratio (EAR), Head Pose Estimation, and Blink Rate.
 - Implemented CLAHE-based image preprocessing for improved facial-feature detection under low-light conditions and a **2-of-3 multi-cue decision** rule to reduce false alarms.
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Certifications

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| • Full Stack Development
Academy : AlphaFly | Nov - Dec 2024 |
| • Power BI
Certified by LOGICPIES | Mar - Apr 2025 |
| • AI Fluency :
Framework & Foundations
Certified by : Anthropic | March 2026 |
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Languages

Tamil | English | Telugu

Achievements & Recognitions

- **2nd Prize – Paper Presentation**, National Symposium, Thiagarajar University. (2024)
 - **3rd Prize – ADAS**, National level Hackathon at ANCIT
 - **2nd Prize – Zonal Level Basketball** Championship. (2021)
 - **1st Prize – District Level Basketball** Championship. (2020)
 - Selected to represent **NCC** at the Republic Day Camp (RDC), (Jan 2019)
participating in the March Past at the **Red Fort, New Delhi**.
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